

2.1.2 Biological molecules

The cells of all living organisms are composed of biological molecules. Proteins, carbohydrates and lipids are three of the key groups of biological

macromolecules that are essential for life. A study of the structure of these macromolecules allows a better understanding of their functions in living organisms.

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	Where appropriate, this section should include diagrams to represent molecular structure and bonding.
(a) how hydrogen bonding occurs between water molecules, and relate this, and other properties of water, to the roles of water for living organisms	A range of roles that relate to the properties of water, including solvent, transport medium, coolant and as a habitat AND roles illustrated using examples of prokaryotes and eukaryotes. HSW2, HSW8
(b) the concept of monomers and polymers and the importance of condensation and hydrolysis reactions in a range of biological molecules	
(c) the chemical elements that make up biological molecules	To include: C, H and O for carbohydrates C, H and O for lipids C, H, O, N and S for proteins C, H, O, N and P for nucleic acids
(d) the ring structure and properties of glucose as an example of a hexose monosaccharide and the structure of ribose as an example of a pentose monosaccharide	To include the structural difference between an α - and a β -glucose molecule AND the difference between a hexose and a pentose monosaccharide.
(e) the synthesis and breakdown of a disaccharide and polysaccharide by the formation and breakage of glycosidic bonds	To include the disaccharides sucrose, lactose and maltose.
(f) the structure of starch (amylose and amylopectin), glycogen and cellulose molecules	HSW8
(g) how the structures and properties of glucose, starch, glycogen and cellulose molecules relate to their functions in living organisms	HSW2, HSW8
(h) the structure of a triglyceride and a phospholipid as examples of macromolecules	To include an outline of saturated and unsaturated fatty acids.
(i) the synthesis and breakdown of triglycerides by the formation (esterification) and breakage of ester bonds between fatty acids and glycerol	

(j) how the properties of triglyceride, phospholipid and cholesterol molecules relate to their functions in living organisms	To include hydrophobic and hydrophilic regions and energy content AND illustrated using examples of prokaryotes and eukaryotes. HSW2, HSW8
(k) the general structure of an amino acid	
(l) the synthesis and breakdown of dipeptides and polypeptides, by the formation and breakage of peptide bonds	
(m) the levels of protein structure	To include primary, secondary, tertiary and quaternary structure AND hydrogen bonding, hydrophobic and hydrophilic interactions, disulfide bonds and ionic bonds. HSW8
(n) the structure and function of globular proteins including a conjugated protein	To include haemoglobin as an example of a conjugated protein (globular protein with a prosthetic group), a named enzyme and insulin. An opportunity to use computer modelling to investigate the levels of protein structure within the molecule. PAG10
(o) the properties and functions of fibrous proteins	To include collagen, keratin and elastin (no details of structure are required).
(p) the key inorganic ions that are involved in biological processes	To include the correct chemical symbols for the following cations and anions: cations: calcium ions (Ca^{2+}), sodium ions (Na^+), potassium ions (K^+), hydrogen ions (H^+), ammonium ions (NH_4^+) anions: nitrate (NO_3^-), hydrogencarbonate (HCO_3^-), chloride (Cl^-), phosphate (PO_4^{3-}), hydroxide, (OH^-).
(q) how to carry out and interpret the results of the following chemical tests: • biuret test for proteins • Benedict's test for reducing and non-reducing sugars • reagent test strips for reducing sugars • iodine test for starch • emulsion test for lipids	PAG9 HSW3, HSW4, HSW5

- (r) quantitative methods to determine the concentration of a chemical substance in a solution
- To include colorimetry and the use of biosensors (an outline only of the mechanism is required).
PAG5
 HSW3, HSW4, HSW5
- (s) (i) the principles and uses of paper and thin layer chromatography to separate biological molecules / compounds
- To include calculation of retention (*R_f*) values.

$$R_f = \frac{\text{distance moved by the solute}}{\text{distance moved by the solvent}}$$
- (ii) practical investigations to analyse biological solutions using paper or thin layer chromatography.
- For example the separation of proteins, carbohydrates, vitamins or nucleic acids.
M0.1, M0.2, M1.1, M1.3, M2.2, M2.3, M2.4
PAG6
 HSW2, HSW3, HSW4